HASHMATH SHAIK

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RESEARCH INTERESTS

Ph.D. researcher and Applied Scientist focused on reliable reasoning in large language models, agentic and tool-augmented AI systems, neuro-symbolic methods, and multimodal representation learning. Delivered production-scale ML systems at Amazon Science, including graph-based fraud detection (projected $80M annual impact) and multi-modal document understanding pipelines, combining rigorous experimentation with measurable real-world outcomes.

EDUCATION

PhD in Computer Engineering, Stony Brook University, **GPA**: 3.89/4.0 — 08/2024 – Present

- **Research Focus:** *Neuro-symbolic and tool-augmented large language models for reliable reasoning, with an emphasis on verification-aware prompting, reinforcement learning for tool-use policies, and agentic collaboration in open-ended coding and problem-solving tasks.*

M.Sc. in Computer Science, Lakehead University, Canada, **GPA**: 4.00/4.0 — 09/2022 – 05/2024

SKILLS

Programming	Python, R, Java, C++, JavaScript, SQL
LLMs & Reasoning	Agentic Systems, RAG, Prompt Optimization, Function Calling
Model Training & Adaptation	Fine-tuning (LoRA, QLoRA), Instruction Tuning, Model Evaluation
Frameworks & Libraries	PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-Learn, SpaCy
Multimodal & Speech	ASR (Whisper), CoreNLP, Vision-Language Models (Qwen2.5-VL)
MLOps & Deployment	Git, Docker, Linux, SageMaker, MLflow, Apache Airflow, Bedrock
Data & Knowledge Systems	MySQL, PostgreSQL, Neo4j, Vector Databases (Pinecone), GraphRAG

PUBLICATIONS

- Villuri, G., Shaik, H., & Doboli, A. Dynamic Adaptation of the LLM Context for Generating Routines with Coupled Semantics. *International Conference on Artificial Neural Networks,* Italy. **(ICANN 2026)**
- Shaik, H., Villuri, G., Doboli, S., & Doboli, A. Concept Combinations with Generator and Validator Agents Prompted Using Insights from Concept Networks. In *Complex Networks & Their Applications XIII*, Lecture Notes in Computer Science, pp. 164–174. **(Complex Networks 2025)**
- Shaik, H., Villuri, G., & Doboli, A. Two Large Language Model-based Methods to Validate Open-Ended Problem Solving in Teams. In *International Conference on Foundation and Large Language Models.* **(FLLM 2025)**
- Shaik, H., & Doboli, A. Using a Symbolic Knowledge Graph to Address LLM Limitations in Analog Circuit Topology Generation. In *IEEE 15th Annual Computing and Communication Workshop and Conference,* Las Vegas. **(CCWC 2025)**
- Villuri, G., Shaik, H., Doboli, A., & Doboli, S. A Stacked Multi-Layered Perceptron–LLM Model for Extracting Relations in Textual Descriptions. In *IEEE Symposium on Computational Intelligence,* Norway. **(SSCI 2025)**

RESEARCH EXPERIENCE

Graduate Research Assistant – Stony Brook University (NY, USA) — 08/2024 – Present

- Built LLM-based generator–validator agent pipelines for collaborative coding and open-ended problem solving, enabling static and real-time multi-agent evaluation of reasoning and implementation correctness.
- Designed neuro-symbolic dialogue and code representations grounded in operational and axiomatic semantics, combined with symbolic knowledge graphs to constrain and validate LLM outputs in analog circuit topology generation.

- Developed tool-augmented, cognitive-architecture-style LLM systems for team-based open-ended problem solving, integrating concept networks and structured validation to reduce hallucinated or unsupported reasoning steps.
- Led research on reliable LLM reasoning through validation, symbolic structure, and agentic workflows, resulting in multiple peer-reviewed publications (MAKE, Complex Networks, FLLM, CCWC, Systems).

Graduate Research Assistant – Mitacs (ON, Canada) 05/2023 – 04/2024

- Developed a clinical question-answering system for PICO-format queries using a two-stage bootstrapping approach with patch attention, achieving 89% answer relevance and outperforming prior baselines by 4.2 F1 points on clinical evidence extraction.
- Deployed a pilot at Thunder Bay Regional Hospital serving 12 clinicians, cutting literature-review time by 60% and improving answer precision by 11% across three clinical domains.

PROFESSIONAL EXPERIENCE

Applied Scientist Intern – Amazon Science (CA, USA) 06/2026 – Present

- Researched graph representation learning methods for fraud detection across Amazon's order network, formulating the problem as similarity search over learned embeddings to match new activity to previously identified fraud rings.
- Designed the experimentation pipeline end to end: constructing training datasets from graph-structured order data, developing a custom labeling strategy that expanded usable training data by 3×, and training and evaluating graph learning models on AWS SageMaker GPU infrastructure.
- Conducted the production readiness analysis characterizing deployment thresholds and model performance trade-offs, informing a go/no-go decision projected to protect approximately $80M in annual fraud losses.

Applied Scientist Intern – Amazon Science (CA, USA) 09/2025 – 01/2026

- Researched multimodal document understanding for automated police report analysis, combining visual and textual reasoning to jointly model document layout, text, and visual features.
- Designed and evaluated a Retrieval-Augmented Generation (RAG) approach that replaced a KNN similarity baseline, improving classification precision from 71% to 80% and recall from 41% to 75% by combining semantic retrieval with LLM-based reasoning.
- Investigated document representation learning through vision-language embeddings, task-aware joint embeddings, and multi-vector representations that preserve layout and typographic structure, advancing the system toward Amazon's 90% production precision target.

AI Software Developer – Aurora Constellations (ON, Canada) 05/2023 – 04/2024

- Designed and developed a healthcare analytics platform integrating LLMs (LLaMA2, GPT) with scalable RESTful APIs via AWS API Gateway, achieving a 40% reduction in network overhead.
- Engineered data analytics pipelines using Python, Hadoop, YARN, and PySpark, and built responsive front-end interfaces improving query efficiency through optimized Hive queries.

Machine Learning Engineer – Tata Consultancy Services (Client: Deutsche Bank) 08/2021 – 08/2022

- Built core modules of an international banking transaction platform in Java (Spring), including the funds transfer page with secure transaction workflows, multi-currency support, and input validation.
- Designed an NLP-based request triaging mechanism using word2vec and GloVe embeddings, reducing manual triage effort and resolution time by 45%, and engineered supporting data pipelines with Apache Airflow, PySpark, and MLflow for transaction monitoring and anomaly detection.

AWARDS AND GRANTS

- **Best Presentation Award, IEEE CCWC, USA** – for "Using a Symbolic Knowledge Graph to Address LLM Limitations in Analog Circuit Topology Generation" 2025
- **AI Research Grant – MITACS, Canada** – $25,000 CAD for LLM-driven, PICO-based evidence medicine research 2024